

SUMMARY OF THE INVENTION

Described herein are systems, devices, and methods for privacy-preserving measurement and interpretation of intimate physiology using a modular, multi-node architecture. In various embodiments, the system comprises a plurality of sensing nodes and an associated computing system configured to receive data from the nodes, derive abstract metrics from raw signals, and present outputs that emphasize longitudinal patterns rather than isolated snapshots.

In certain embodiments, the system includes multiple device types comprising at least one wearable node configured for extended measurement, at least one session-based node configured for baseline acquisition, and at least one context node configured to measure physiologic signals that support interpretation of intimate-region patterns. The nodes may measure one or more modalities including circumferential response, pressure, strain, temperature, hydration or moisture proxies, impedance, motion, activity, posture, respiratory proxies, heart rate, heart rate variability, tissue-adjacent signals, and combinations thereof.

The computing system may transform signals into derived features and privacy-preserving metrics including indices, ratios, stability measures, duration measures, trend measures, pattern curves, temporal signatures, variability signatures, and confidence indicators. In various embodiments, outputs are presented using non-anatomical abstractions comprising bars, curves, ranges, zones, scores, or other non-explicit representations.

In some embodiments, the computing system performs cross-device correlation and contextual interpretation, in which intimate-region signals are analyzed in view of one or more of baseline acquisition results, cardiophysiologic context signals, activity context, and within-user historical patterns, thereby improving interpretability and reducing ambiguity. In certain embodiments, the system further supports privacy-preserving correlation across different users, including correlation between a first user's metrics and a second user's metrics, across a range of pairing contexts, without requiring explicit anatomical imagery or explicit labeling.

In various embodiments, the system supports multiple workflows including baseline establishment, longitudinal tracking, context-tagged interpretation, and correlation across devices and time. The described architecture enables expansion to additional nodes and modalities while maintaining privacy-preserving output formats and standardized correlation methods.